



National University of Sciences and Technology (NUST)
School of Electrical Engineering and Computer Science

Department of Computing

Course Code: SE 200

Class: BSCS-13E

Software Engineering

Instructor: Sir Sarosh Tahir

Project Deliverable 1

Team Members

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1. Problem Statement:

Mental health has turned into one of the biggest public health challenges of our time, and honestly, the numbers are kind of alarming. According to the World Health Organization (WHO), over 1 billion people worldwide are currently living with some form of mental health disorder. Anxiety and depression alone make up more than two-thirds of all these cases, which shows just how widespread these issues really are. On top of that, depression and anxiety cost the global economy roughly \$1 trillion every single year in lost productivity. So it is not just a health issue, it is an economic one too.

What makes things worse is that most people who need help are not getting it. The WHO reports that in low-income countries, fewer than 10% of people affected by mental health issues actually receive any kind of care. Even in wealthier nations, only about one-third of people with depression get proper treatment. The reasons are pretty clear: there are not enough mental health professionals, therapy is expensive, there is still a lot of stigma attached to seeking help, and people simply do not have easy access to support when they need it most.

Young adults are hit especially hard. Nearly half of all mental health disorders start before age 18, and the 20 to 29 age group has seen the largest increase in prevalence over the past decade. Students and working professionals are constantly dealing with academic pressure, career stress, social media comparisons, and burnout, but they rarely have the time or money to sit down with a therapist on a regular basis.

This is where the gap becomes really obvious. People need a simple, private, and affordable way to keep track of their emotional health on a daily basis. They need something that helps them spot patterns in their mood, understand what is triggering their stress, and gives them practical suggestions to feel better. Right now, traditional mental health services are failing to meet this demand because of cost, availability, and social barriers.

Moodify aims to solve this problem by providing a mobile-based mood tracking and mental wellness app that makes it easy for users to log their emotions, track daily activities, get personalized insights, and access helpful suggestions. The whole idea is to make mental health self-care feel as routine and natural as checking the weather on your phone. By putting the right tools in people's hands, Moodify wants to bridge the gap between needing help and actually getting it, especially for younger users who are comfortable with technology but may not be comfortable walking into a therapist's office.



2. Objectives:

The following objectives define what Moodify will aim to achieve. Each one is specific, measurable, and tied to a real outcome so we can actually track whether the app is doing what it is supposed to do.

Objective 1: Build a Quick and Intuitive Mood Logging System

Design a mood entry flow that takes no more than 30 seconds to complete, with options to select mood level, emotions, activities, and an optional note. The target is for at least 70% of active users to log their mood at least 5 days a week within the first month of use. If logging feels like a chore, people will stop doing it, so simplicity is the priority here.

Objective 2: Deliver Actionable Mood Insights

Develop an analytics dashboard that shows users their mood trends over time (weekly, monthly, yearly) and correlates mood with tracked activities like sleep, exercise, and social interactions. The goal is for 60% of users to report that Moodify helped them identify at least one mood pattern or trigger within 30 days of regular use, measured through in-app surveys.

Objective 3: Provide Personalized Wellness Suggestions

Build a recommendation engine that offers context-aware suggestions such as breathing exercises, journaling prompts, or activity changes based on mood patterns. The target is for at least 50% of users who receive suggestions to engage with at least one recommended activity per week.

Objective 4: Achieve Strong User Retention

Reach a 30-day retention rate of at least 40% after launch, which is above the industry average for health and fitness apps (which typically sits around 25 to 30%). This will be measured through analytics tracking of daily and monthly active users.

Objective 5: Ensure Data Privacy and User Trust

Implement end-to-end encryption for all mood and personal data, allow offline-first local storage, and provide full data export and deletion options. The goal is to achieve a user trust score of at least 4.2 out of 5 in post-onboarding surveys, because people will not share their emotions with an app they do not trust.



3. Stakeholders & Users:

Any project like this involves a bunch of different people and groups who all have a stake in how the app turns out. Here is a breakdown of who is involved and what they care about.

3.1 Primary Users:

End Users (People Tracking Their Mood)

These are the main users of the app. They include college students dealing with academic stress, working professionals managing burnout, and anyone who wants to understand their emotional patterns better. Most of them are between the ages of 16 and 35, are comfortable using smartphones, and are looking for something quick and private. They care about ease of use, helpful insights, and knowing their data is safe.

Therapists and Counselors

Mental health professionals who might recommend Moodify to their clients as a supplementary tool between sessions. They want to see reliable mood data that can inform treatment, and they care about clinical validity and the ability for clients to share reports or summaries.

3.2 Secondary Stakeholders:

Product Development Team

This includes the designers, developers, and QA testers building the app. They care about clear requirements, feasible timelines, and building something that actually solves the problem without overcomplicating things.

Project Manager / Scrum Master

The person keeping everything on track. They coordinate between the team, make sure milestones are hit, and handle blockers. Their main concern is delivering the MVP on time and within scope.

University or Healthcare Partners

Organizations like university counseling centers or workplace wellness programs that might integrate or recommend Moodify to their communities. They care about data privacy compliance, evidence-based features, and accessibility standards.

3.3 Business Stakeholders:

Investors / Sponsors

People funding the project who care about market viability, user growth, retention rates, and a clear path to monetization (like freemium subscriptions). They want to see measurable KPIs and competitive positioning..



4. Competitive Analysis

To understand where Moodify fits in the market, we analyzed two of the most well-known mood tracking apps: Daylio and Moodfit. We pulled real market data from Sensor Tower to get an accurate picture of how these apps are actually performing in terms of downloads, revenue, and rankings. Here is what we found.

4.1 Competitor 1: Daylio

Daylio is one of the most popular mood tracking apps out there, available on both iOS and Android. It is often described as a micro-journal because the whole point is that you can log your mood and activities without having to type anything. You just tap an emoji for your mood, select what activities you did, and optionally add a short note. The app then crunches this data and shows you charts, statistics, and a colorful Year in Pixels view that maps out your entire year by mood color.

Daylio has a 4.8-star rating on both the Apple App Store and Google Play Store, with hundreds of thousands of reviews. The free version covers the basics well, but the premium version (\$4.99/month or \$35.99/year) unlocks advanced statistics, unlimited goals, automatic backups, and removes ads on Android. Data is stored locally on the device and backed up to Google Drive or iCloud.

Sensor Tower Data for Daylio (Source: Sensor Tower)

Metric	Daylio
App Name	Daylio Journal - Mood Tracker
Developer	Habitics
App ID	net.daylio
Category	Lifestyle
Price	Free (with in-app purchases)
Install Range	10M - 50M
Downloads (Last Month, Worldwide)	100K
Revenue (Last Month, Worldwide)	\$50K
Category Ranking (US)	#262 Lifestyle - Downloads



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Top Countries / Regions	Brazil, Russia, US
Content Rating	Everyone
Current Version	1.64.1
Last Updated	2026/03/27
Publisher Country	Slovakia
Country Release Date	2015/08/17
Worldwide Release Date	2015/08/17
Minimum OS Version	8.0

Strengths:

- Super simple and fast to use. Logging takes just a few taps.
- Highly customizable moods, icons, activities, and color themes.
- Excellent analytics with mood charts, correlations, and Year in Pixels.
- Strong privacy approach with local data storage.
- Massive user base (10M to 50M installs) proving strong product-market fit.
- Goal tracking and daily streak system to keep users engaged.

Weaknesses:

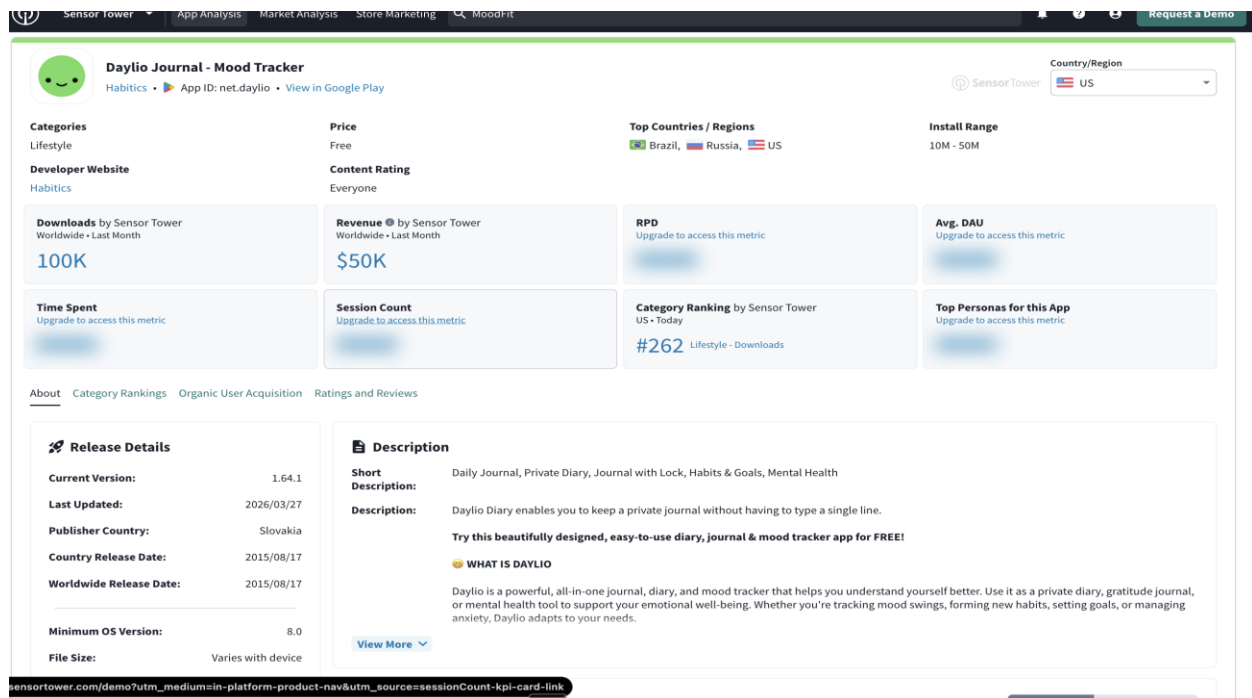
- No CBT tools, breathing exercises, or guided wellness content.
- No clinical assessments like PHQ-9 or GAD-7.
- Limited journaling depth since it is designed for minimal writing.
- No web version or desktop compatibility.
- Does not offer AI-driven personalized suggestions.



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Screenshot From Sensor Tower:





4.2 Competitor 2: Moodfit:

Moodfit positions itself as a mental fitness app rather than just a mood tracker. It goes well beyond simple mood logging by including CBT-based thought records, breathing exercises, mindfulness meditation, gratitude journaling, medication tracking, and even a nervous system regulation tool. It has been recognized by CNET as one of the best mental health apps in 2024 and 2025, and Verywell Mind named it the best mood tracker in 2023.

Moodfit is free to download with basic features, but the premium version unlocks the full toolkit including CBT thought records, advanced breathing exercises, and nervous system tracking. It also includes validated clinical assessments like the PHQ-9 for depression and GAD-7 for anxiety, which is a big deal for users working with therapists. The app tracks how activities like sleep, exercise, and nutrition correlate with mood over time.

Sensor Tower Data for Moodfit (Source: Sensor Tower)

Metric	Moodfit
App Name	Moodfit: Mental Health Fitness
Developer	Roble Ridge Software LLC
App ID	com.robleridge.Moodfit
Category	Medical
Price	Free (with in-app purchases)
Install Range	100K - 500K
Downloads (Last Month, Worldwide)	< 5K
Revenue (Last Month, Worldwide)	< \$5K
Category Ranking (US)	N/A
Top Countries / Regions	US, United Kingdom, India
Content Rating	Everyone



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Current Version	3.4
Last Updated	2026/02/10
Publisher Country	US
Country Release Date	2017/01/04
Worldwide Release Date	2017/01/04
Minimum OS Version	6.0

Strengths:

- Comprehensive mental health toolkit covering CBT, breathwork, mindfulness, and more.
- Clinically validated assessments (PHQ-9, GAD-7) for tracking depression and anxiety.
- Medication and custom activity tracking to see what actually helps.
- Exportable mood and medication charts to share with therapists.
- Educational content on topics like rumination, procrastination, and motivation.

Weaknesses:

- Very low downloads (under 5K per month) despite being on the market since 2017, suggesting weak discoverability.
- Revenue under \$5K per month shows the monetization model is not working well.
- No category ranking on Sensor Tower (N/A), meaning it is not competitive in store visibility.
- The free version is quite limited, pushing users toward the premium plan.
- The interface can feel overwhelming because there are so many features packed in.
- Some users have reported bugs like freezing and issues syncing data.
- Lacks the fun, gamified elements (streaks, achievements, Year in Pixels) that keep casual users engaged.



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Screenshot From Sensor Tower:

**Moodfit: Mental Health Fitness**
Roble Ridge Software LLC • App ID: com.robleridge.Moodfit • View in Google Play

Country/Region
Sensor Tower US

Categories
Medical

Price
Free

Top Countries / Regions
US, United Kingdom, India

Install Range
100K - 500K

Developer Website
Roble Ridge Software LLC

Content Rating
Everyone

Downloads by Sensor Tower
Worldwide • Last Month
< 5k

Revenue by Sensor Tower
Worldwide • Last Month
< \$5k

RPD
Upgrade to access this metric

Avg. DAU
Upgrade to access this metric

Time Spent
Upgrade to access this metric

Session Count
Upgrade to access this metric

Category Ranking by Sensor Tower
Today
N/A

Top Personas for this App
Upgrade to access this metric

[About](#) [Category Rankings](#) [Organic User Acquisition](#) [Ratings and Reviews](#)

Release Details
Current Version: 3.4
Last Updated: 2026/02/10
Publisher Country: US
Country Release Date: 2017/01/04
Worldwide Release Date: 2017/01/04
Minimum OS Version: 6.0
File Size: Varies with device

Description
Short Description: Reduce stress & build resilience - mood tracking, CBT, breathwork & mindfulness
Description: Moodfit is a powerful mood tracker and mental fitness app designed to help you reduce stress, practice CBT, and build lasting habits with breathwork, mindfulness, and a nervous system tool.

Award-Winning Mental Health App
• Best Mental Health Apps of 2024 & 2025 - CNET
• Best Mood Tracker (2023) - Verywell Mind
• Best Overall Mental Health App (2020-2022) - Verywell Mind
• Featured in Forbes, Vogue, Men's Health, PsychCentral, and more

[View More](#)



4.3 Feature Comparison Table:

The table below compares Daylio, Moodfit, and our proposed Moodify app across the key features that matter most to users in this space.

Feature	Daylio	Moodfit	Moodify (Ours)
Quick Mood Logging	Yes (icon-based)	Yes	Yes (icon + voice)
Activity Tracking	Yes (customizable)	Yes	Yes (customizable)
Mood Charts & Analytics	Advanced	Moderate	Advanced + AI insights
Year in Pixels / Calendar	Yes	No	Yes (planned)
CBT Tools / Thought Records	No	Yes	Yes
Breathing Exercises	No	Yes (premium)	Yes (free)
Mindfulness / Meditation	No	Yes	Yes
Gratitude Journal	No	Yes	Yes
Clinical Assessments (PHQ-9, GAD-7)	No	Yes	Yes
Medication Tracking	No	Yes	Planned (v2)
AI Personalized Suggestions	No	No	Yes
Nervous System Tool	No	Yes	Planned (v2)
Gamification (Streaks, Achievements)	Yes	No	Yes
Data Export (PDF/CSV)	Yes (premium)	Yes	Yes (free)



Therapist Sharing / Reports	No	Yes (export charts)	Yes (dedicated reports)
Offline Support	Yes (local storage)	Partial	Yes (offline-first)
Privacy & Local Storage	Yes	Cloud-based	Yes (local + encrypted)
Pricing (Premium)	\$4.99/mo or \$35.99/yr	\$9.99/mo or \$39.99/yr	Freemium TBD

4.4 Where Moodify Fits In:

Looking at both the feature comparison and the Sensor Tower data, there is a clear gap in the market. Daylio dominates in terms of user base and engagement because it nails the basics: it is fast, fun, and simple. But it completely lacks therapeutic tools like CBT, breathing exercises, and clinical assessments. Moodfit has all the clinical depth you could want, but it is clearly struggling to grow its user base, pull in revenue, or even rank in the app store.

Moodify aims to sit right in the middle. It combines the simplicity and engagement of Daylio (quick logging, streaks, Year in Pixels) with the clinical depth of Moodfit (CBT tools, assessments, therapist reports), while adding a layer of AI-powered personalization that neither competitor currently offers. The goal is to make mental health tracking feel approachable and even enjoyable, while still providing genuinely useful tools for people who need more than just a mood chart.

By keeping core features free, prioritizing privacy with local-first storage and encryption, designing specifically for users aged 16 to 35, and investing in strong app store optimization from day one (to avoid the visibility problem Moodfit has), Moodify is positioned to fill a real gap in the current mental health app landscape.



5. Project Management Tools:

For managing the Moodify project, we set up two dedicated platforms: Notion for documentation and knowledge management, and Linear for task tracking and sprint planning. Both tools are being actively used by the team to keep everything organized, transparent, and easy to access throughout the development lifecycle.

5.1 Notion Workspace:

We created a Notion workspace called "Moodify" to serve as the central hub for all project documentation. The workspace is organized under a main page titled "Moodify - Project Wiki" which contains the project vision, problem statement, key barriers, target user personas, and all research materials in one place.

What the Notion Workspace Contains:

- Vision statement summarizing the core idea behind Moodify: making mental health self-care feel as routine as checking the weather.
- Problem statement with key WHO statistics and the treatment gap that Moodify addresses.
- Key barriers section highlighting the main obstacles to mental health support (cost, stigma, limited access, workforce shortages).
- Why young adults are hit hardest section with data on age-related prevalence increases.
- Target users table with four detailed personas (College Student, Young Professional, Therapy-Adjacent User, Self-Improvement Seeker), each with descriptions and core needs.

The Notion workspace is shared with all team members and the instructor for full transparency. It will be continuously updated as we move through sprints, adding design artifacts, meeting notes, and sprint retrospectives.



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Screenshot of Notion Workspace:

Vision

Make mental health self-care feel as routine and natural as checking the weather on your phone.

Problem statement

Over 1 billion people worldwide live with a mental health disorder (WHO). Fewer than 10% in low-income countries receive any care. Even in wealthy nations, only about one-third of people with depression get proper treatment.

Key barriers

- Not enough professionals
- Therapy is expensive
- Stigma persists
- Access is limited

Why young adults are hit hardest

- Nearly half of all mental health disorders start before age 18.
- Ages 20–29 have seen the largest increase in prevalence.
- Students and working professionals face academic pressure, career stress, social media comparisons, and burnout.

Moodify exists to fill this gap.

Target users

Persona	Description	Core need
College student	Ages 18–24, academic stress, social pressure, first time living independently	Quick daily check-in, pattern recognition, free wellness tools
Young professional	Ages 24–35, career stress, burnout, work-life balance struggles	Insights into what affects mood, actionable suggestions
Therapy-adjacent user	Any age, currently seeing or considering a therapist	Clinical assessments, therapist reports, medication tracking
Self-improvement seeker	Wellness-oriented, tracks fitness and diet, wants to add mental health	Gamification, streaks, gratitude journaling, holistic view



5.2 Linear Workspace

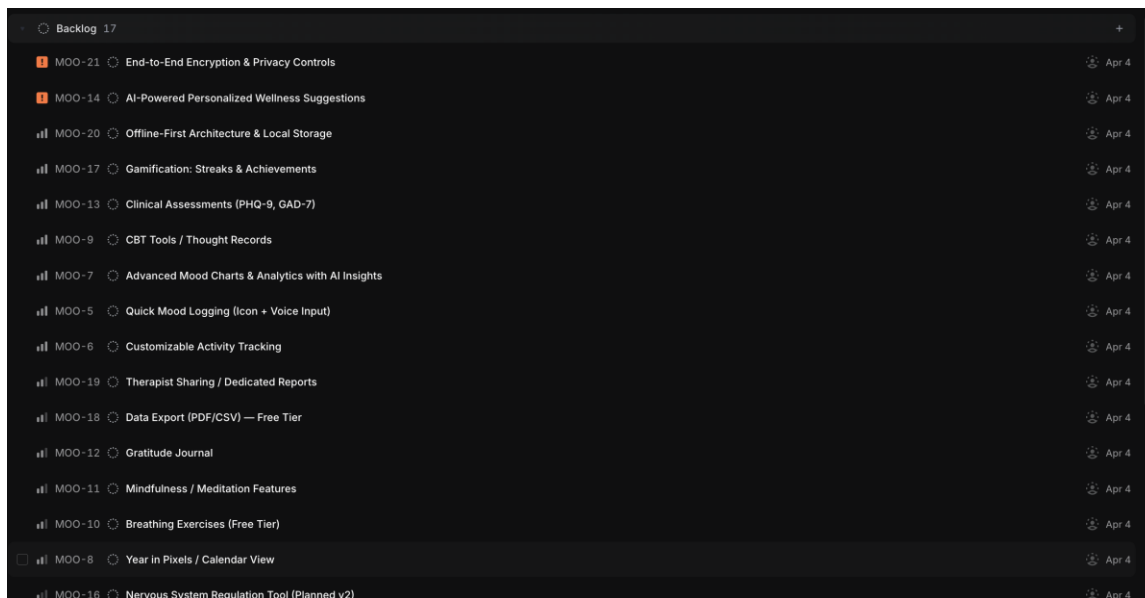
For task tracking and sprint management, we chose Linear because it is fast, clean, and built specifically for software teams. We set up a Linear workspace called "Moodify" where all features, tasks, and issues are tracked using Linear's issue system with unique IDs (MOO-1, MOO-2, etc.).

How We Organized the Linear Workspace:

We broke down all the features from our competitive analysis and feature comparison table into individual issues in Linear. Each issue represents a specific feature or task that needs to be built, and is tagged with a priority level and status. This gives us a clear backlog to pull from during sprint planning.

Backlog Overview:

The backlog currently contains 17 feature items, each mapped directly to our feature comparison table and project objectives. Some of the key items include: Quick Mood Logging (MOO-5), Advanced Mood Charts with AI Insights (MOO-7), CBT Tools and Thought Records (MOO-9), Clinical Assessments for PHQ-9 and GAD-7 (MOO-13), AI-Powered Personalized Wellness Suggestions (MOO-14), Gamification with Streaks and Achievements (MOO-17), and End-to-End Encryption with Privacy Controls (MOO-21). Each item is prioritized so the team knows what to focus on first when sprint planning begins.



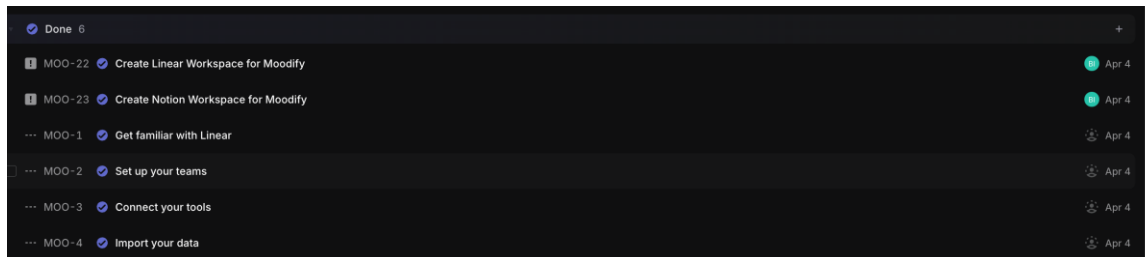


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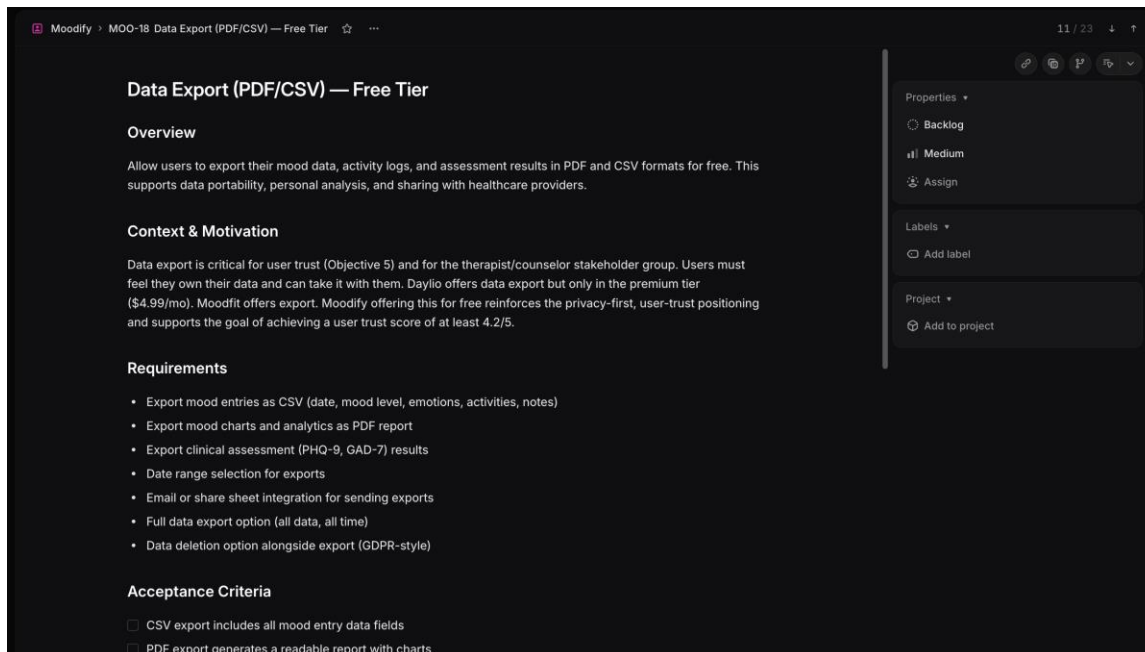
Completed Tasks:

We have already completed 6 initial setup tasks, which include getting familiar with Linear (MOO-1), setting up teams (MOO-2), connecting tools (MOO-3), importing data (MOO-4), and most importantly, creating both the Linear workspace (MOO-22) and the Notion workspace (MOO-23) for Moodify. These are marked as Done, showing that the project infrastructure is fully in place.



Issue Detail View:

Each issue in Linear is not just a title. We have written detailed descriptions for every feature item that include an overview explaining what the feature does, a context and motivation section linking back to our project objectives and competitive analysis, a requirements list with specific functional needs, and acceptance criteria that define when the feature is considered complete. This level of detail ensures that when a developer picks up a task, they know exactly what to build and how to verify it works. Below is an example showing the MOO-18 (Data Export) issue, which includes all four sections.





Team Members:

Members

Q Search by name or email

All

Export CSV

Invite

Name 4	Email	Status	Teams	Joined	Last seen
Active 4					
araza.bscs23seecs@seecs.edu.pk araza.bscs23seecs	araza.bscs23seecs@seecs.edu.pk	Admin	1 team	Apr 4	Apr 4
Bilal bilalrananust351	bilalrananust351@gmail.com	Admin	1 team	Apr 4	Apr 4
mchaudry.bscs23seecs@seecs.edu.pk mchaudry.bscs23seecs	mchaudry.bscs23seecs@seecs.edu.pk	Admin	1 team	Apr 4	Apr 4
Muhammad Samama Usman musman.bscs23seecs	musman.bscs23seecs@seecs.edu.pk	Admin	1 team	Apr 4	Online

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